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Track: FinTech | Challenge ID: FT-01

Design and Develop an Intelligent Personal Finance Management Platform for Expense Analytics, Budget Optimization, and Savings Recommendations

Problem Description

Individuals frequently use multiple payment modes and receive fragmented transaction records, making it difficult to understand spending behaviour, maintain realistic budgets, and build disciplined savings habits. The challenge is to develop a personal finance platform that converts transaction data into actionable financial insight rather than functioning as a simple expense-entry application. The system should accept manually entered or simulated transaction records, classify expenses into meaningful categories, compare actual spending against user-defined budgets, identify unusual or excessive spending patterns, and provide explainable recommendations for improving monthly financial discipline. The design must separate presentation, business logic, and data storage so that the prototype demonstrates sound software development practice. Real bank credentials or production payment data must not be required; teams should work with synthetic or user-created test data.

Technical Scope and Minimum Requirements

The prototype should implement secure user access, transaction capture/import, rule-based or model-assisted expense categorization, category-wise and time-wise analytics, configurable budgets, savings goals, and an explainable recommendation engine. At least one analytical technique should be visible in the implementation, such as budget variance analysis, trend detection, recurring-expense identification, or a lightweight spending forecast. The interface should provide meaningful charts or indicators and should handle invalid, duplicate, and missing transaction inputs. A persistent data store and a documented API or modular service layer are expected.

Expected Outcomes

A successful solution will allow a user to view current financial position, understand where money is being spent, receive early warnings when budgets are likely to be exceeded, and track progress toward one or more savings goals. Judges should be able to verify that recommendations are produced from defined logic or data patterns rather than static text. The system should also produce a concise monthly summary containing total income, total expense, category distribution, budget variance, savings achieved, and recommended actions.

12-Hour Minimum Deliverable and Live Demonstration

Within 12 hours, teams should deliver a working end-to-end prototype using synthetic transactions for at least one month. The live demonstration must show data entry/import, automatic or assisted categorization, dashboard analytics, a budget alert, a savings recommendation, and persistence of the generated records. Advanced forecasting, bank integration, or production-grade financial advice may be treated as optional extensions rather than mandatory components.